

ZIHUI LU

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TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, JavaScript (ES6+), HTML5/CSS3, Tailwind CSS
Backend & Data: Node.js, Express, REST APIs, JWT, PostgreSQL, MongoDB, SQLite, Supabase
Tools & Testing: Git, Docker, Jenkins (CI/CD), Nginx, Jira, Vitest
Other Languages: C#/.NET, C/C++ (STL, Valgrind), Python (PyTorch, NumPy)

PROFESSIONAL EXPERIENCE

- Shenzhen LaitCloud Technology** | *Software Developer Intern* Jan 2026 – June 2026
- Built and deployed a production **multi-tenant SaaS** for construction finance tracking (**React/TypeScript** frontend, **Node.js/Express + MongoDB** backend) with **database-per-tenant** isolation, serving real users. [↗ Live Demo](#)
 - Implemented **JWT** authentication, bcrypt-hashed credentials, and **role-based access control** across super-admin / distributor / tenant roles, and deployed it on a self-managed cloud server with **Docker** and **Nginx**.
 - Produced promotional video content using **CapCut Pro**, AI agents, **Remotion**, and **Hyperframe**.
- SOTI Inc.** | *Software Developer Intern* May 2022 – Aug 2023
- Designed a **C#** tool using **Octokit SDK** to auto-generate PRs for managed components, reducing manual steps by **90%** and saving **3+ hours per task**.
 - Integrated the tool into a **Jenkins** CI/CD pipeline, supporting the team's release workflow.
 - Contributed to a **C#** service tracking **500+** BDD tests in real-time, reducing failure diagnosis time from **2 hours to 30 minutes**.

EDUCATION

University of Toronto Graduated May 2025
B.Eng in Computer Engineering (Minor: Artificial Intelligence & Business)
Key courses: Data Structures & Algorithms, Computer Networks, Databases, Machine Learning

TECHNICAL PROJECTS

- Simple Skill Manager** [↗ GitHub](#) (*Next.js, React, TypeScript, SQLite*) Summer 2026
- Built a local full-stack skill-library app with **Next.js 15 (App Router)**, **React**, **TypeScript**, and **SQLite** (better-sqlite3), published as an **npm package**.
 - Implemented **content-hash deduplication** and per-agent **symlink/copy management** with realpath safety checks across **13** auto-detected AI coding agents.
 - Added a marketplace, **GitHub backup/sync**, an MCP-config viewer, and a bilingual (English/Chinese) UI, with core logic covered by **Vitest** tests.
- School Food Program Dashboard (U of T)** [↗ Live Demo](#) (*Next.js, TypeScript, React*) Fall 2024 - Spring 2025
- Led a team to optimize frontend performance by **78%** (PageSpeed 49 → 87) using **code-splitting** and **SSR**.
 - Built **Node.js/Express** REST API endpoints serving nutrition data to **1,000+** monthly users.
 - Developed interactive data visualizations (DataWrapper & Flourish) to highlight regional nutrition trends.
- Toronto GIS Map Engine** (*C++, EZGL, Valgrind*) Spring 2021
- Developed a **high-performance** map renderer using **Dijkstra's algorithm** (95% query success rate).
 - Reduced memory leaks by **70%** through Valgrind profiling and smart pointer refactoring.

ADDITIONAL

- Languages:** English, Mandarin, Cantonese
- Leadership:** Tutored 30+ high school students in STEM subjects, improving grades by 20% on average.
- Technical Writing:** Maintain detailed project documentation